

AVIONICS
PROFESSIONAL ELECTIVES-III

VII Semester										
Course Code		Category		Hours / Week			Credits	Maximum Marks		
A6AE44		PEC		L	T	P	C	CIE	SEE	Total
				3	0	0	3	40	60	100
UNIT-I		AVIONICS TECHNOLOGY								
Evolution of electronics. Nature of microelectronic devices- processors, memory devices. Introduction to avionics- systems integration– need- data bus systems – MIL STD 1553 bus system, ARINC 429 / ARINC 629 bus systems, optical data bus systems. Integrated modular avionics architectures										
UNIT-II		AIRCRAFT INSTRUMENTATION - SENSORS AND DISPLAYS								
Air data sensors, magnetic sensing, inertial sensing, radar sensors. The electromechanical instrumented flight deck, early flight deck instruments, attitude direction indicator, horizontal situation indicator, altimeter, airspeed indicator. Advanced flight deck display system architectures, display systems, display media, future flight deck displays										
UNIT-III		COMMUNICATION AND NAVIGATION								
Radio frequency spectrum, communication systems, HF, VHF, satellite communications; ATC transponder, traffic collision avoidance system. Navigational aids. Automatic Direction Finding, VHF Omni Range, Distance Measuring Equipment; TACAN, VORTAC. Satellite navigation systems – the GPS. Instrument landing system, transponder landing system, microwave landing system. Hyperbolic navigation systems										
UNIT-IV		NAVIGATION SYSTEMS								
Basic navigation, radio, inertial navigations, satellite navigation- GPS, differential GPS, wide area augmentation systems, local area augmentation system, GPS overlay programme. Integrated navigation, sensor usage. Flight management system (FMS). FMS control										
UNIT-V		AIRBORNE RADAR, ASTRIONICS								
Propagation of Radar waves- functional elements of radar- antenna- transmitter; Types of Radar- Pulse Doppler- civil aviation applications, military applications; Attitude determination & control of spacecraft- Magnetometers, sun sensors, star trackers and telemetry systems										
Text Books:										
1. Moir, I. and Seabridge, A., Civil Avionics Systems, AIAA Education Series, AIAA, 2002, ISBN 56347589-8. 2. Henderson, M. F., Aircraft Instruments & Avionics for A & P Technicians, Jeppesen Sanderson Training Products, 1993, ISBN 0-89100-422-X										
Reference Books:										
1. Helfrick, A., Principles of Avionics, Avionics Communications Inc. Leesburg, 2000, VA 20177, USA, ISBN 1-885544-10-3. 2. Moir, I. and Seabridge, A., Aircraft Systems: Mechanical, Electrical and Avionics Subsystems Integration, AIAA Education Series, AIAA, 2001, ISBN 1-56347506-5.										
COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:										
1. Illustrates the architecture of the avionics in the airplane 2. Explain the flight deck display systems used in the aircraft 3. Describe the communication and navigation systems 4. Discuss the Navigation and flight management systems 5. Explain the telemeter systems used in the space craft										